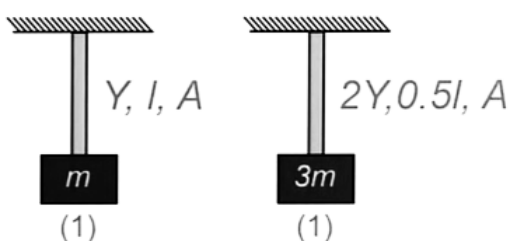


JEE-Main-04-04-2026 (Memory Based)

[EVENING SHIFT]

Physics

Question: Two strings with length (l_1, l_2), Young's moduli (Y_1, Y_2) are elongated under two weights as shown. Find $\Delta l_1 / \Delta l_2$.



Options:

- (a) 3
- (b) $2/3$
- (c) $4/3$
- (d) $1/3$

Answer: (c)

Solution:

Step-by-Step Calculation

1. **Elongation of String 1 (Δl_1):**

$$\Delta l_1 = \frac{(mg) \cdot l}{A \cdot Y}$$

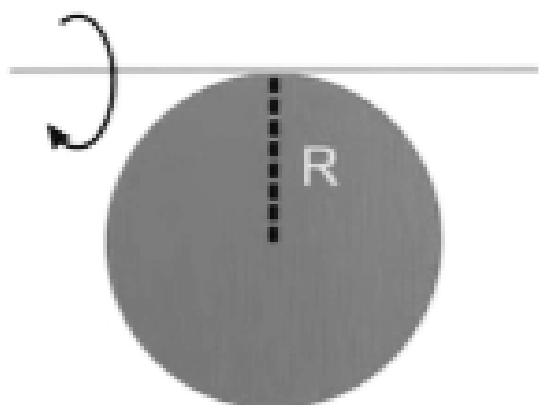
2. **Elongation of String 2 (Δl_2):**

$$\Delta l_2 = \frac{(3mg) \cdot (0.5l)}{A \cdot (2Y)} = \frac{1.5 \cdot mg \cdot l}{2 \cdot A \cdot Y} = \frac{3}{4} \left(\frac{mg \cdot l}{A \cdot Y} \right)$$

3. **Ratio ($\Delta l_1 / \Delta l_2$):**

$$\frac{\Delta l_1}{\Delta l_2} = \frac{1}{3/4} = \frac{4}{3}$$

Question: Consider a ring of radius R , rotates about a horizontal axis as shown. Time period of small oscillation



Options:

(a) $2\pi\sqrt{\frac{5R}{g}}$

(b) $2\pi\sqrt{\frac{R}{2g}}$

(c) $2\pi\sqrt{\frac{3R}{2g}}$

(d) $2\pi\sqrt{\frac{R}{g}}$

Answer: (c)

Solution:

Step-by-Step Calculation

1. Elongation of String 1 (Δl_1):

$$\Delta l_1 = \frac{(mg) \cdot l}{A \cdot Y}$$

2. Elongation of String 2 (Δl_2):

$$\Delta l_2 = \frac{(3mg) \cdot (0.5l)}{A \cdot (2Y)} = \frac{1.5 \cdot mg \cdot l}{2 \cdot A \cdot Y} = \frac{3}{4} \left(\frac{mg \cdot l}{A \cdot Y} \right)$$

3. Ratio ($\Delta l_1 / \Delta l_2$):

$$\frac{\Delta l_1}{\Delta l_2} = \frac{1}{3/4} = \frac{4}{3}$$

Step-by-Step Calculation

1. Elongation of String 1 (Δl_1):

$$\Delta l_1 = \frac{(mg) \cdot l}{A \cdot Y}$$

2. Elongation of String 2 (Δl_2):

$$\Delta l_2 = \frac{(3mg) \cdot (0.5l)}{A \cdot (2Y)} = \frac{1.5 \cdot mg \cdot l}{2 \cdot A \cdot Y} = \frac{3}{4} \left(\frac{mg \cdot l}{A \cdot Y} \right)$$

3. Ratio ($\Delta l_1 / \Delta l_2$):

$$\frac{\Delta l_1}{\Delta l_2} = \frac{1}{3/4} = \frac{4}{3}$$

Question: Two Cars A & B are moving on a road with speed 100 km/h and 80 km/h a stone is thrown from Car B with speed V km/h relative to it. Stone hit the Car A with speed 5 m/s w.r.t Car A. (ignore gravity.). The V is _____

Options:

- (a) 18
- (b) 38
- (c) 48
- (d) 20

Answer: (b)

Solution:

Step-by-Step Calculation

1. **Elongation of String 1 (Δl_1):**

$$\Delta l_1 = \frac{(mg) \cdot l}{A \cdot Y}$$

2. **Elongation of String 2 (Δl_2):**

$$\Delta l_2 = \frac{(3mg) \cdot (0.5l)}{A \cdot (2Y)} = \frac{1.5 \cdot mg \cdot l}{2 \cdot A \cdot Y} = \frac{3}{4} \left(\frac{mg \cdot l}{A \cdot Y} \right)$$

3. **Ratio ($\Delta l_1 / \Delta l_2$):**

$$\frac{\Delta l_1}{\Delta l_2} = \frac{1}{3/4} = \frac{4}{3}$$

Step-by-Step Calculation

1. **Elongation of String 1 (Δl_1):**

$$\Delta l_1 = \frac{(mg) \cdot l}{A \cdot Y}$$

2. **Elongation of String 2 (Δl_2):**

$$\Delta l_2 = \frac{(3mg) \cdot (0.5l)}{A \cdot (2Y)} = \frac{1.5 \cdot mg \cdot l}{2 \cdot A \cdot Y} = \frac{3}{4} \left(\frac{mg \cdot l}{A \cdot Y} \right)$$

3. **Ratio ($\Delta l_1 / \Delta l_2$):**

$$\frac{\Delta l_1}{\Delta l_2} = \frac{1}{3/4} = \frac{4}{3}$$

Step-by-Step Calculation

1. Elongation of String 1 (Δl_1):

$$\Delta l_1 = \frac{(mg) \cdot l}{A \cdot Y}$$

2. Elongation of String 2 (Δl_2):

$$\Delta l_2 = \frac{(3mg) \cdot (0.5l)}{A \cdot (2Y)} = \frac{1.5 \cdot mg \cdot l}{2 \cdot A \cdot Y} = \frac{3}{4} \left(\frac{mg \cdot l}{A \cdot Y} \right)$$

3. Ratio ($\Delta l_1 / \Delta l_2$):

$$\frac{\Delta l_1}{\Delta l_2} = \frac{1}{3/4} = \frac{4}{3}$$

4. Solving for V

Now substitute the km/h values into the relative velocity equation:

$$18 = (V + 80) - 100$$

$$18 = V - 20$$

$$V = 18 + 20$$

$$V = 38 \text{ km/h}$$

Question: List-I presents some physical quantity and list-II presents the dimensions. Match the two list appropriately.

	List-I		List-II
(A)	f = work function	(1)	$ML^2T^{-3}I^{-1}$
(B)	V_s = stopping potential	(2)	$M^{\circ}L^{\circ}T^{-1}$
(C)	h = Planck's constant	(3)	ML^2T^{-2}
(D)	f = frequency	(4)	ML^2T^{-1}

Options:

(a) A→(3); B→(4); C→(2); D→(1)

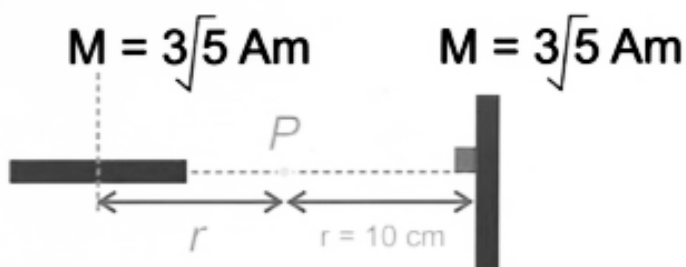
(b) A→(3); B→(1); C→(4); D→(2)

(c) A→(2); B→(3); C→(4); D→(1)

(d) A→(4); B→(3); C→(1); D→(2)

Answer: (b)

Question: Point P at $r = 10$ cm distance from centers of two bar magnet each of magnetic moment $3\sqrt{5}$ Am. Magnetic field at P is



Options:

(a) 1.5 mT

(b) 2 mT

(c) 2.5 mT

(d) 4.5 mT

Answer: (a)

Solution:

1. Given Data

- **Magnetic Moment (M):** $3\sqrt{5} \text{ Am}^2$ for each magnet.
- **Distance (r):** $10 \text{ cm} = 0.1 \text{ m}$.
- **Configuration:** * Point P is on the **axial line** of the horizontal magnet.
 - Point P is on the **equatorial line** of the vertical magnet.

2. Magnetic Field Formulas

- **Axial Field (B_a):** $\frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$
- **Equatorial Field (B_e):** $\frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}$

Note: $\frac{\mu_0}{4\pi} = 10^{-7} \text{ T} \cdot \text{m/A}$

3. Step-by-Step Calculation

Field due to Horizontal Magnet (B_1):

$$B_1 = 10^{-7} \times \frac{2 \times 3\sqrt{5}}{(0.1)^3} = 10^{-7} \times \frac{6\sqrt{5}}{10^{-3}} = 6\sqrt{5} \times 10^{-4} \text{ T}$$

Field due to Vertical Magnet (B_2):

$$B_2 = 10^{-7} \times \frac{3\sqrt{5}}{(0.1)^3} = 3\sqrt{5} \times 10^{-4} \text{ T}$$

Net Magnetic Field (B_{net}):

Since the fields are perpendicular to each other (B_1 is horizontal, B_2 is vertical):

3. Step-by-Step Calculation

Field due to Horizontal Magnet (B_1):

$$B_1 = 10^{-7} \times \frac{2 \times 3\sqrt{5}}{(0.1)^3} = 10^{-7} \times \frac{6\sqrt{5}}{10^{-3}} = 6\sqrt{5} \times 10^{-4} \text{ T}$$

Field due to Vertical Magnet (B_2):

$$B_2 = 10^{-7} \times \frac{3\sqrt{5}}{(0.1)^3} = 3\sqrt{5} \times 10^{-4} \text{ T}$$

Net Magnetic Field (B_{net}):

Since the fields are perpendicular to each other (B_1 is horizontal, B_2 is vertical):

Question: Bullets are fired from a gun in all the direction at all angles. Maximum range on horizontal ground is 6.4 m. Find speed of bullet at the time of firing.

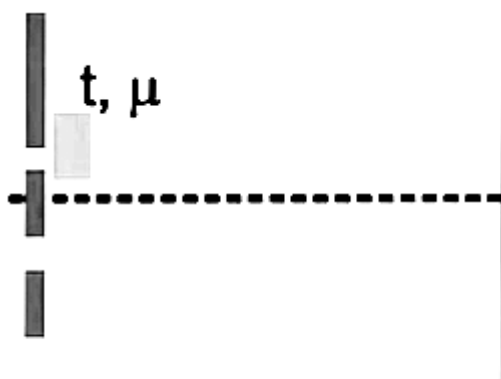
Options:

- (a) $4\sqrt{2}$ m/s
- (b) 8 m/s
- (c) 16 m/s
- (d) $8\sqrt{2}$ m/s

Answer: (d)

Solution:

Question: In YDSE initially pattern was observed without Covering the slits. Then after. one transparent sheet of thickness t and refractive index $\mu = 3/2$ is being placed before one of the slit. It is Observed that 1st maxima with transparent sheet coincides with 7th maxima without the sheet. Find the value of t .



Options:

(a) $t = 12\lambda$

(b) $t = 6\lambda$

(c) $t = 9\lambda$

(d) $t = 3\lambda$

Answer: (a)

Solution:



1. Given Data

- **Refractive Index (μ):** $3/2$ (or 1.5)
- **Condition:** The **1st maxima** with the sheet coincides with the **7th maxima** of the original pattern (without the sheet).
- **Wavelength:** λ

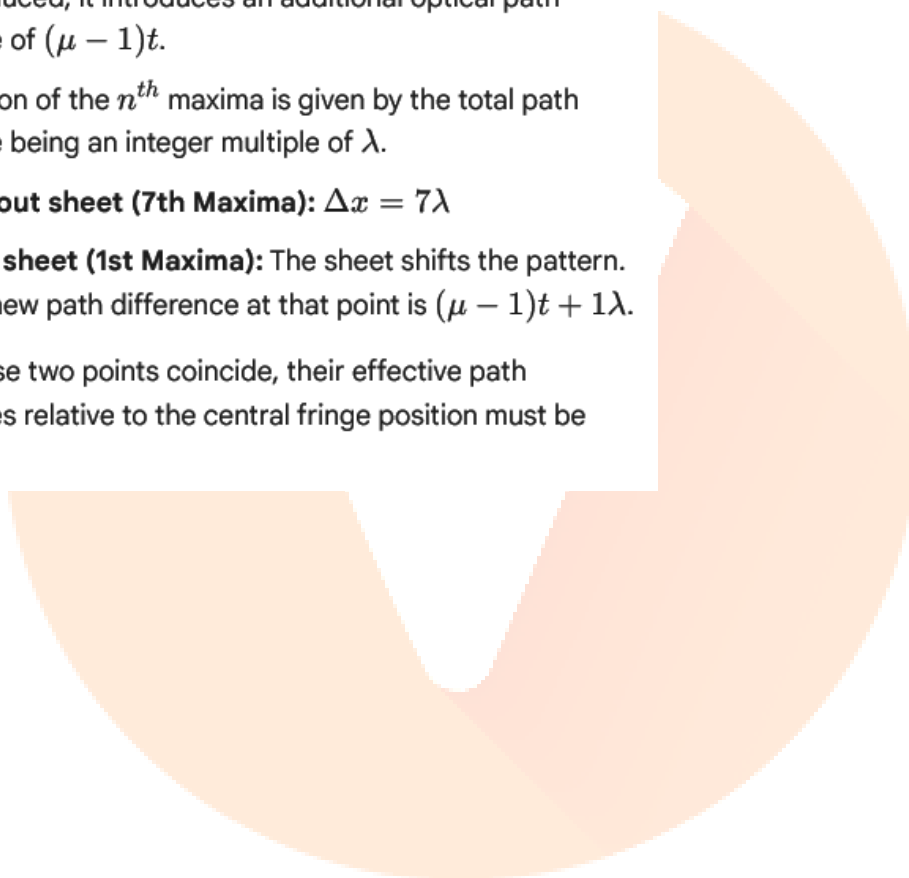
2. Concept: Path Difference & Shift

When a transparent sheet of thickness t and refractive index μ is introduced, it introduces an additional optical path difference of $(\mu - 1)t$.

The position of the n^{th} maxima is given by the total path difference being an integer multiple of λ .

- **Without sheet (7th Maxima):** $\Delta x = 7\lambda$
- **With sheet (1st Maxima):** The sheet shifts the pattern.
The new path difference at that point is $(\mu - 1)t + 1\lambda$.

Since these two points coincide, their effective path differences relative to the central fringe position must be equal:



$$(\mu - 1)t + 1\lambda = 7\lambda$$

3. Step-by-Step Calculation

1. Simplify the equation:

$$(\mu - 1)t = 7\lambda - 1\lambda$$

$$(\mu - 1)t = 6\lambda$$

2. Substitute $\mu = 3/2$:

$$\left(\frac{3}{2} - 1\right)t = 6\lambda$$

$$\left(\frac{1}{2}\right)t = 6\lambda$$

3. Solve for t :

$$t = 2 \times 6\lambda$$

$$t = 12\lambda$$

Question: For $\vec{F} = (5\hat{i} + 2\hat{j})N$ as a mass of 0.1 kg, initially rest at origin, acting for 2 sec. gives final position as
Options:

- (a) $4\hat{i} + 10\hat{j}$
- (b) $7\hat{i} + 80\hat{j}$
- (c) $100\hat{i} + 40\hat{j}$
- (d) $50\hat{i} + 20\hat{j}$

Answer: (c)

Solution:

1. Given Data

- **Force (F):** $(5\hat{i} + 2\hat{j})$ N
 - **Mass (m):** 0.1 kg
 - **Time (t):** 2 sec
 - **Initial State:** At rest ($u = 0$) at the origin $(0, 0)$.
-

2. Step-by-Step Calculation

Step A: Find Acceleration (a)

Using Newton's Second Law ($F = ma$):

$$a = \frac{F}{m} = \frac{5\hat{i} + 2\hat{j}}{0.1}$$

$$a = (50\hat{i} + 20\hat{j}) \text{ m/s}^2$$

Step B: Find Final Position (s)

Using the equation of motion $s = ut + \frac{1}{2}at^2$:

Since $u = 0$:

$$s = \frac{1}{2} \cdot (50\hat{i} + 20\hat{j}) \cdot (2)^2$$

$$s = \frac{1}{2} \cdot (50\hat{i} + 20\hat{j}) \cdot 4$$

$$s = 2 \cdot (50\hat{i} + 20\hat{j})$$

$$s = 100\hat{i} + 40\hat{j}$$

Question: Two scenarios of a dipole present in two electric field

$$(i) \vec{E}_1 = E_0 \hat{x} \quad (ii) \vec{E}_2 = E_0 \hat{x} + 2E_0 \hat{y} + 2E_0 \hat{z}.$$

The dipole is in stable equilibrium in separation instants. Time periods T_1 & T_2 are in ratio of (for small oscillations)

Options:

(a) $\sqrt{3}$

(b) $3^{\frac{1}{8}}$

(c) $5^{\frac{1}{2}}$

(d) $5^{\frac{1}{4}}$

Answer: (d)

Solution:

1. Given Data

- **Case 1:** $E_1 = E_0 \hat{i}$
 - Magnitude $E_1 = E_0$
- **Case 2:** $E_2 = E_0 \hat{i} + 2E_0 \hat{j} + 2E_0 \hat{k}$
 - Magnitude $E_2 = \sqrt{E_0^2 + (2E_0)^2 + (2E_0)^2}$
 - $E_2 = \sqrt{E_0^2 + 4E_0^2 + 4E_0^2} = \sqrt{9E_0^2} = 3E_0$

2. Formula for Time Period (T)

For a dipole undergoing small angular oscillations in a uniform electric field, the time period is:

$$T = 2\pi \sqrt{\frac{I}{pE}}$$

1. Given Data

- **Case 1:** $E_1 = E_0 \hat{i}$
 - Magnitude $E_1 = E_0$
- **Case 2:** $E_2 = E_0 \hat{i} + 2E_0 \hat{j} + 2E_0 \hat{k}$
 - Magnitude $E_2 = \sqrt{E_0^2 + (2E_0)^2 + (2E_0)^2}$
 - $E_2 = \sqrt{E_0^2 + 4E_0^2 + 4E_0^2} = \sqrt{9E_0^2} = 3E_0$

2. Formula for Time Period (T)

For a dipole undergoing small angular oscillations in a uniform electric field, the time period is:

$$T = 2\pi \sqrt{\frac{I}{pE}}$$

Question: Electron, proton are accelerated with same potential to achieve de-Broglie

wavelength of λ_1 and λ_2 ($m_p = 1849 m_e$) then $\frac{\lambda_1}{\lambda_2}$ is

Options:

- (a) 37
- (b) 43
- (c) 1/41
- (d) 1/48

Answer: (b)

Solution:

Question: The position of the particle changes as function of time as $x = 54t$; $y = 15t^2 - 4t$. Find velocity vector angle with x- axis at $t = 2$ sec.

Options:

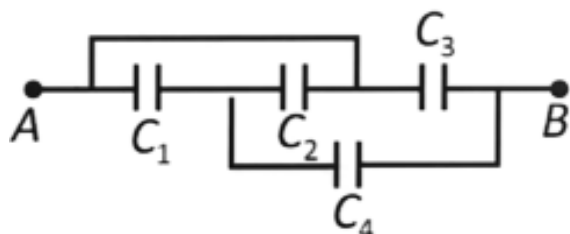
- (a) $\theta = \tan^{-1} \left(\frac{13}{17} \right)$
- (b) $\theta = \tan^{-1} \left(\frac{13}{15} \right)$
- (c) $\theta = \tan^{-1} \left(\frac{25}{27} \right)$

$$\theta = \tan^{-1} \left(\frac{28}{27} \right)$$

(d)

Answer: (d)

Question: Find equivalent capacitance between points A & B. $C_1 = C_2 = C_3 = 1\mu\text{F}$, $C_4 = 2\mu\text{F}$



Options:

- (a) $6\mu\text{F}$
- (b) $2\mu\text{F}$
- (c) $4/3\mu\text{F}$
- (d) $5\mu\text{F}$

Answer: (b)

Question: Assertion : For two samples of O_2 and H_2 , V_{rms} are same if temperature is the same.

Reason : Average molar kinetic energy of H_2 and O_2 are same if their temperature are the same.

Options:

- (a) A is correct, R is correct explanation of A
- (b) A is correct, R is correct but not correct explanation
- (c) A is correct, R is incorrect
- (d) A is incorrect, R is correct

Answer: (d)

Question: Magnetic field component of an EM wave is given as

$$\vec{B} = \beta_0 \sin(2\pi vt - kx) \hat{j}.$$

Find electric field component of the EM wave

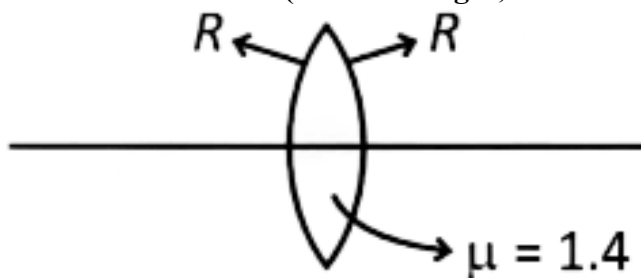
Options:

- (a) $\frac{\lambda\beta_0 V}{2} \cos(2\pi vt - kx) \hat{k}$
- (b) $2\lambda\beta_0 v \sin(2\pi vt - kx) \hat{k}$
- (c) $\frac{\lambda\beta_0 v}{2} \sin(2\pi vt - kx) \hat{i}$
- (d) $\lambda\beta_0 v \sin(2\pi vt - kx)$

Answer: (d)

Question: A biconvex lens is made up of material of refractive index $\mu = 1.4$ as shown.

Find the ratio of $\frac{f}{R}$. (f = focal length, R = curvature radius)

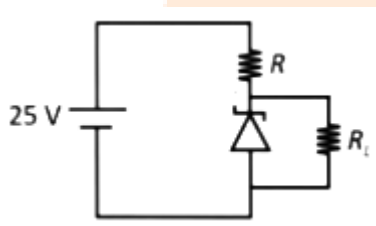


Options:

- (a) $3/4$
- (b) $3/5$
- (c) $5/3$
- (d) $5/4$

Answer: (d)

Question: In the given circuit Zener breakdown voltage is 10 V if power loss in resistance R is 0.5 W. Find value of R.



Options:

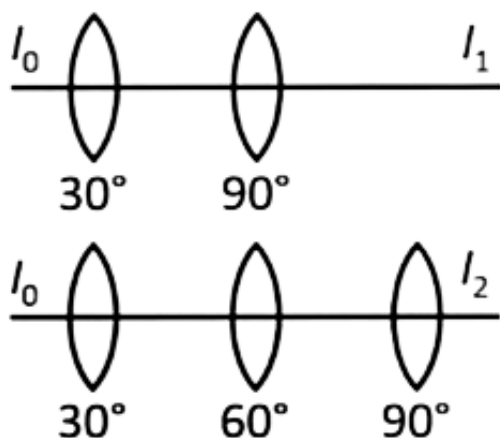
- (a) 125Ω
- (b) 450Ω
- (c) 350Ω
- (d) 250Ω

Answer: (b)

Solution:

Question: Set of polarisers is given below. Unpolarised light falls with intensity I_0 . Find

value of $\frac{I_1}{I_2}$. Angles marked are with reference line.



Options:

- (a) 8/9
- (b) 4/9
- (c) 9/3
- (d) 3/4

Answer: (b)

Question: A current carrying coil is oriented in a way that axis of the coil is at 30° with existing magnetic field $\beta = 0.4$ T. Number of turn of the coil is $N = 125$ and current $I = 1$ Ampere is flowing through it. Area of the coil is 2 cm^2 . Find torque on the coil.

Options:

- (a) 0.05 m
- (b) 0.02 N-m
- (c) 0.005 N-m
- (d) 0.05 N-m

Answer: (c)

Question: At what height h from the surface of earth, gravitational field is $\frac{1}{9}$ th that at gravitational field at surface then $\frac{h}{R}$ is

Answer: (2)

Question: An ideal liquid is flowing through a pipe having different cross-section as shown.



$E = 15 \text{ mm}$; $r_2 = 30 \text{ mm}$; $V_2 = 50 \text{ cm/s}$.

Find v_1 (in /s).

Answer: (2)

Solution:

Question: The magnetising field intensity in a region having magnetic field of 4 Tesla

and relative permeability in the region is $\mu_r = 20$ is $\frac{10^7}{x\pi}$ (A/m). Find the value of X.

Answer: (20)



JEE-Main-04-04-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: Manganese forms oxide and fluoride with highest oxidation state. The difference in the highest oxidation state of Mn in the oxide and fluoride is

Options:

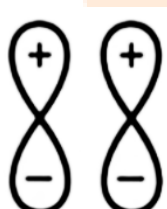
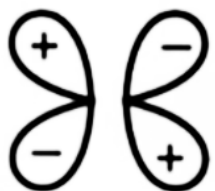
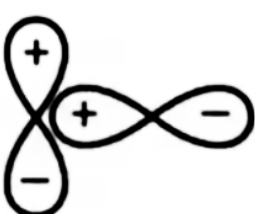
- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (c)

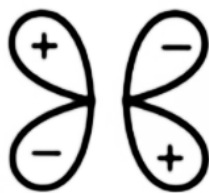
Solution: (c) 3

Question: Two atom are bonding along Z-axis. How will π^* orbital look like: (Z is internuclear axis)

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (c)



Solution: (c)

Question: Match the list-I with list-II

List-I (Name)	List-II (Reagent used)
(I) Tollen's reagent	(P) $\text{Cu}^{2+} / \text{OH}^-$
(II) Fehling's reagent	(Q) $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$
(III) Williamson method	(R) dil. KMnO_4
(IV) Bayer's reagent	(S) $\text{R-X} + \text{RO}^- \text{Na}$

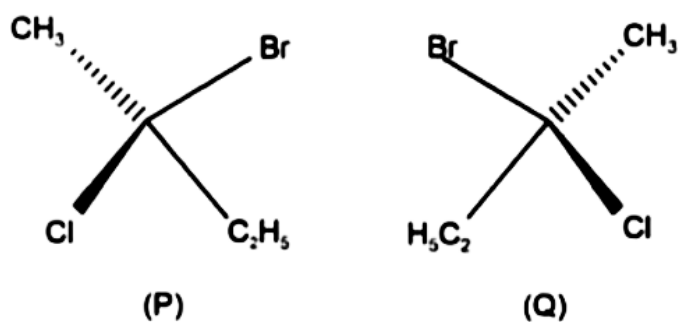
Options:

- (a) (I)-(Q), (II)-(P), (III)-(S), (IV)-(R)
 (b) (I)-(P), (II)-(Q), (III)-(R), (IV)-(S)
 (c) (I)-(Q), (II)-(S), (III)-(P), (IV)-(R)
 (d) (I)-(Q), (II)-(R), (III)-(P), (IV)-(S)

Answer: (a)

Solution: (a) (I)-(Q), (II)-(P), (III)-(S), (IV)-(R)

Question: Identify the relation between the following structure (P) and (Q)



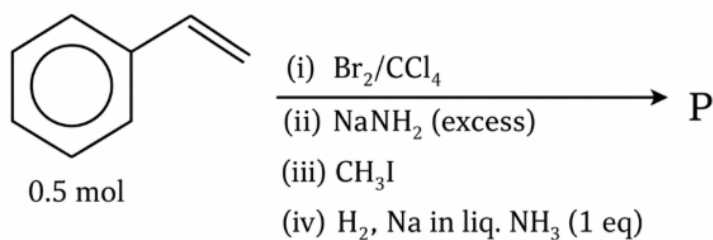
Options:

- (a) Homomers
 (b) Enantiomers
 (c) Diastereomers
 (d) None of these

Answer: (b)

Solution: (b) Enantiomers

Question: Consider the reaction:



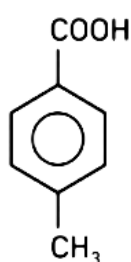
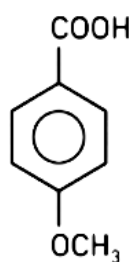
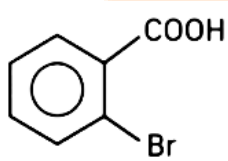
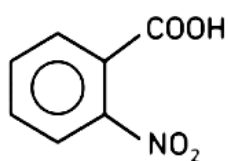
Find the mass of 'P' obtained in 'g'. 'P' is an aromatic compound

Options:

Answer: (59)

Solution:

Question: Compare acidic strength of following molecule



Options:

(a) $A > B > C > D$

(b) $A > B > D > C$

(c) $B > A > C > D$

(d) $B > A > D > C$

Answer: (b)

Solution: (b) $A > B > D > C$

Question: Consider the following complex ions:

A. $[\text{Fe}(\text{CN})_6]^{3-}$

B. $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

C. $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$

D. $[\text{FeCl}_6]^{3-}$

Which of the following options correctly represents the order of the number of unpaired electrons?

Options:

(a) $A < B = C = D$

(b) $A < C < B = D$

(c) $A < C < B < D$

(d) $A = C < B = D$

Answer: (a)

Solution: (a) $A < B = C = D$

Question: Acetic acid is mixed with NaOH in a 5 L container in the molar ratio 4 : 1. When the ratio changes to 3 : 1, the change in pH of the solution is (find the magnitude of change).

Given: $\log 3 = 0.48$, $\log 2 = 0.30$

Options:

- (a) 0.39
- (b) 0.36
- (c) 0.18
- (d) 0.78

Answer: (c)

Solution: (c) 0.18

Question: If the first ionisation energy of Mg is 738 kJ/mol, then the second ionisation energy of Mg will be

(Given $\Rightarrow \text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$, $\Delta H = +2189$ kJ/mol)

Options:

- (a) 1451
- (b) 1683
- (c) 1476
- (d) 2927

Answer: (a)

Solution: (a) 1451

Question: One of the following ions will precipitate in presence of H_2S and NH_4OH : Fe^{3+} , Mn^{2+} , Cu^{2+} , Pb^{2+}

The highest possible oxidation state of that species is

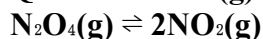
Options:

- (a) (+2)
- (b) (+4)
- (c) (+7)
- (d) (+6)

Answer: (c)

Solution: (c) (+7)

Question: $\text{N}_2\text{O}_4(\text{g})$ is dissociated to $\text{NO}_2(\text{g})$ according to the reaction:



At an equilibrium pressure of 1 atm, N_2O_4 is 50% dissociated. Temperature is 50°C . Calculate ΔG° for the given reaction.

Given: $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$, $\ln(4/3) = 0.3$

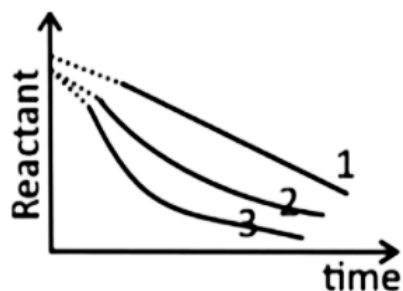
Options:

- (a) -384.67 J/mol
- (b) -804.27 J/mol
- (c) -124.63 J/mol
- (d) -834.56 J/mol

Answer: (b)

Solution: (b) -804.27 J/mol

Question: Consider the following graphical relations:



Correct statement among the following are

Options:

- (a) (1) depicts first order reaction
- (b) (2) and (3) depict second order reactions
- (c) (2) and (3) depict first order reaction with same rate constant
- (d) (2) and (3) depict first order reactions, having rate constant for (2) is less than that of (3)

Answer: (d)

Solution: (d) (2) and (3) depict first order reactions, having rate constant for (2) is less than that of (3)

JEE-Main-04-04-2026 (Memory Based)
[EVENING SHIFT]
Maths

Question: If the system of equations

$$x + y + z = 5$$

$$x + 2y + 3z = 9$$

$x + 3y + \lambda z = \mu$ has infinite number of solutions. Then the value of $\lambda + \mu$ is

Options:

- (a) 18
- (b) 23
- (c) 21
- (d) 22

Answer: (b)

Solution:

1. Set up the Augmented Matrix

We represent the system of equations as a matrix $[A|B]$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 2 & 3 & 9 \\ 1 & 3 & \lambda & \mu \end{array} \right]$$

2. Apply Row Operations

To simplify the matrix, we want to create zeros in the first column below the first row.

- $R_2 \rightarrow R_2 - R_1$
- $R_3 \rightarrow R_3 - R_1$

The matrix becomes:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 2 & 4 \\ 0 & 2 & \lambda - 1 & \mu - 5 \end{array} \right]$$

Next, we create a zero in the second column of the third row:

$$\bullet R_3 \rightarrow R_3 - 2R_2$$

The final row-echelon form is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 5 & \mu - 13 \end{array} \right]$$

3. Condition for Infinite Solutions

For a system to have an **infinite number of solutions**, the last row must consist entirely of zeros. This happens when:

$$1. \lambda - 5 = 0 \implies \lambda = 5$$

$$2. \mu - 13 = 0 \implies \mu = 13$$

4. Final Calculation

The question asks for the value of $\lambda + \mu$:

$$\lambda + \mu = 5 + 13 = 18$$

Question: The number of non-negative integer solution of the equation $a + b + 2c = 22$, is

Options:

- (a) 124
- (b) 144
- (c) 135
- (d) 136

Answer: (b)

Solution:

1. Strategy: Fix the variable with the coefficient

Since c is multiplied by 2, it has the most restricted range. We can rewrite the equation as:

$$a + b = 22 - 2c$$

For a and b to be non-negative integers, $22 - 2c$ must be ≥ 0 . This means c can be any integer from 0 to 11.

2. Calculate solutions for each c

For any equation of the form $a + b = n$, the number of non-negative integer solutions is always $n + 1$. (For example, if $a + b = 2$, the solutions are $(0, 2)$, $(1, 1)$, $(2, 0)$, which is $2 + 1 = 3$ solutions).

Let's list the number of solutions for each value of c :

- If $c = 0$, then $a + b = 22 \implies 22 + 1 = \mathbf{23}$ solutions.
- If $c = 1$, then $a + b = 20 \implies 20 + 1 = \mathbf{21}$ solutions.
- If $c = 2$, then $a + b = 18 \implies 18 + 1 = \mathbf{19}$ solutions.
- ...
- If $c = 11$, then $a + b = 0 \implies 0 + 1 = \mathbf{1}$ solution.

3. Find the Total Sum

The total number of solutions is the sum of this sequence:

$$\text{Total} = 23 + 21 + 19 + \dots + 3 + 1$$

This is the sum of the **first 12 odd numbers**. A quick math shortcut is that the sum of the first n odd numbers is always n^2 .

$$\text{Total} = 12^2 = \mathbf{144}$$

Question: Evaluate $(0.2)^{\log_{\sqrt{5}} \alpha} + (0.04)^{\log_5 \beta}$, **if** $\alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$

$$\beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$$

Options:

- (a) 4
- (b) 8
- (c) 12
- (d) 10

Answer: (b)

Solution:

1. Calculate α and β

Both α and β are infinite geometric series. We use the sum formula $S_{\infty} = \frac{a}{1-r}$, where a is the first term and r is the common ratio.

- **For α :** $a = \frac{1}{4}$ and $r = \frac{1}{2}$.

$$\alpha = \frac{1/4}{1 - 1/2} = \frac{1/4}{1/2} = \frac{1}{2}$$

- **For β :** $a = \frac{1}{3}$ and $r = \frac{1}{3}$.

$$\beta = \frac{1/3}{1 - 1/3} = \frac{1/3}{2/3} = \frac{1}{2}$$

So, we have $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{2}$.

2. Simplify the Expression

The expression is $(0.2)^{\log_{\sqrt{5}} \alpha} + (0.04)^{\log_5 \beta}$.

Evaluate the First Term:

$$(0.2)^{\log_{\sqrt{5}}(1/2)}$$

- Write 0.2 as 5^{-1} and $\sqrt{5}$ as $5^{1/2}$.
- Use the change of base formula: $\log_{b^k} x = \frac{1}{k} \log_b x$.

$$\text{Exponent} = \log_{5^{1/2}}(1/2) = \frac{1}{1/2} \log_5(1/2) = 2 \log_5(1/2) = \log_5(1/2)^2 = \log_5(1/4)$$

- Now substitute back:

$$(5^{-1})^{\log_5(1/4)} = 5^{-\log_5(1/4)} = 5^{\log_5(1/4)^{-1}} = (1/4)^{-1} = 4$$

Evaluate the Second Term:

$$(0.04)^{\log_5(1/2)}$$

- Write 0.04 as $\frac{4}{100} = \frac{1}{25} = 5^{-2}$.

$$(5^{-2})^{\log_5(1/2)} = 5^{-2 \log_5(1/2)} = 5^{\log_5(1/2)^{-2}} = (1/2)^{-2} = 4$$

3. Final Calculation

Summing the two terms:

$$\text{Value} = 4 + 4 = 8$$

Question: In expansion of $\left(9x + \frac{1}{3\sqrt{3}}\right)^{18}$, if the coefficient of term independent of x is 221k, then the value of k is

Options:

- (a) 82
- (b) 83
- (c) 84
- (d) 86

Answer: (c)

Solution:

1. General Term Formula

The general term T_{r+1} in the expansion of $(a + b)^n$ is given by:

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

For the given expression $\left(9x - \frac{1}{3\sqrt{x}}\right)^{18}$:

- $n = 18$
- $a = 9x$
- $b = -\frac{1}{3\sqrt{x}} = -\frac{1}{3}x^{-1/2}$

2. Substitute into the Formula

$$T_{r+1} = \binom{18}{r} (9x)^{18-r} \left(-\frac{1}{3}x^{-1/2}\right)^r$$

Now, let's group the constants and the x terms separately:

$$T_{r+1} = \binom{18}{r} (9)^{18-r} \left(-\frac{1}{3}\right)^r \cdot x^{18-r} \cdot x^{-r/2}$$

$$T_{r+1} = \binom{18}{r} (9)^{18-r} \left(-\frac{1}{3}\right)^r \cdot x^{18-\frac{3r}{2}}$$

3. Find the Term Independent of x

For the term to be independent of x , the exponent of x must be **zero**:

$$18 - \frac{3r}{2} = 0 \implies 36 = 3r \implies r = 12$$

4. Calculate the Coefficient

Substitute $r = 12$ back into the coefficient part of our expression:

$$\text{Coefficient} = \binom{18}{12} (9)^{18-12} \left(-\frac{1}{3}\right)^{12}$$

$$\text{Coefficient} = \binom{18}{6} (3^2)^6 \left(\frac{1}{3^{12}}\right)$$

$$\text{Coefficient} = \binom{18}{6} \cdot 3^{12} \cdot \frac{1}{3^{12}} = \binom{18}{6}$$

Now, let's evaluate $\binom{18}{6}$:

$$\binom{18}{6} = \frac{18 \times 17 \times 16 \times 15 \times 14 \times 13}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 18,564$$

5. Solve for k

We are given that the coefficient is $221k$.

$$221k = 18,564$$

$$k = \frac{18,564}{221} = 84$$

Question: If $\sqrt{3}i, 1$ are the roots of $x^3 + ax^2 + bx + c = 0$ $a, b, c \in \mathbb{R}$. Then

$\int_{-1}^1 (x^3 + ax^2 + bx + c) dx$ is

Options:

(a) $\frac{20}{3}$

(b) $\frac{10}{3}$

(c) $-\frac{20}{3}$

(d) $-\frac{10}{3}$

Answer: (c)

Solution:

1. Identify the Roots

We are given two roots: $x_1 = 1$ and $x_2 = \sqrt{3}i$.

Since the coefficients a , b , and c are **real numbers**, complex roots must occur in **conjugate pairs**. This means that if $\sqrt{3}i$ is a root, its conjugate $-\sqrt{3}i$ must also be a root.

So, the three roots are:

- $x_1 = 1$
- $x_2 = \sqrt{3}i$
- $x_3 = -\sqrt{3}i$

2. Construct the Polynomial

We can write the polynomial as the product of its factors:

$$f(x) = (x - 1)(x - \sqrt{3}i)(x + \sqrt{3}i)$$

Using the identity $(a - b)(a + b) = a^2 - b^2$:

$$f(x) = (x - 1)(x^2 - (\sqrt{3}i)^2)$$

$$f(x) = (x - 1)(x^2 - (-3))$$

$$f(x) = (x - 1)(x^2 + 3)$$

Expanding this gives us:

$$f(x) = x^3 - x^2 + 3x - 3$$

3. Evaluate the Integral

We need to calculate $\int_{-1}^1 (x^3 - x^2 + 3x - 3)dx$.

We can use the property of integrals over a symmetric interval $[-a, a]$:

- The integral of an **odd function** (like x^3 and $3x$) is **0**.

- The integral of an **even function** (like $-x^2$ and -3) is $2 \int_0^a f(x)dx$.

Applying this property:

$$\begin{aligned}\int_{-1}^1 (x^3 - x^2 + 3x - 3)dx &= \int_{-1}^1 (x^3 + 3x)dx + \int_{-1}^1 (-x^2 - 3)dx \\ &= 0 + 2 \int_0^1 (-x^2 - 3)dx\end{aligned}$$

Now, integrate:

$$\begin{aligned}I &= 2 \left[-\frac{x^3}{3} - 3x \right]_0^1 \\ I &= 2 \left[\left(-\frac{1}{3} - 3 \right) - (0) \right] \\ I &= 2 \left[-\frac{10}{3} \right] = -\frac{20}{3}\end{aligned}$$

Question: $\int_{-1}^1 \cot^{-1}(1 + x + x^2)dx$ is equal to

Options:

- (a) $2\tan^{-1}2 - \frac{1}{2}\ln 5$
- (b) $2\tan^{-1}2 + \frac{1}{2}\ln 5$
- (c) $\frac{1}{2}\tan^{-1}2 - \frac{1}{5}\ln 2$
- (d) $\frac{1}{2}\tan^{-1}2 + \frac{1}{5}\ln 2$

Answer: (a)

Solution:

1. Simplify the Integrand

Using the identity $\cot^{-1} u = \tan^{-1} \left(\frac{1}{u} \right)$ (valid here since $x^2 + x + 1$ is always positive), we rewrite the integral:

$$I = \int_{-1}^1 \tan^{-1} \left(\frac{1}{1+x+x^2} \right) dx$$

Next, we manipulate the argument of the tangent function to use the identity

$$\tan^{-1} \left(\frac{a-b}{1+ab} \right) = \tan^{-1} a - \tan^{-1} b:$$

$$\frac{1}{1+x+x^2} = \frac{1}{1+x(1+x)} = \frac{(x+1)-x}{1+x(x+1)}$$

Substituting this back into the integral:

$$I = \int_{-1}^1 [\tan^{-1}(x+1) - \tan^{-1} x] dx$$

2. Split and Evaluate the Integrals

We can split this into two separate integrals:

$$I = \int_{-1}^1 \tan^{-1}(x+1) dx - \int_{-1}^1 \tan^{-1} x dx$$

- **Second Part:** $\tan^{-1} x$ is an **odd function** ($f(-x) = -f(x)$). Since the limits are symmetric about zero (-1 to 1), this integral is **0**.

- **First Part:** Let's use a substitution for $\int_{-1}^1 \tan^{-1}(x+1) dx$.

Let $t = x + 1 \Rightarrow dt = dx$.

When $x = -1$, $t = 0$. When $x = 1$, $t = 2$.

The integral becomes: $I = \int_0^2 \tan^{-1} t dt$

3. Integration by Parts

To evaluate $\int \tan^{-1} t dt$, we use integration by parts: $\int u dv = uv - \int v du$.

$$\text{Let } u = \tan^{-1} t \Rightarrow du = \frac{1}{1+t^2} dt$$

$$\text{Let } dv = dt \Rightarrow v = t$$

$$\int \tan^{-1} t dt = t \tan^{-1} t - \int \frac{t}{1+t^2} dt = t \tan^{-1} t - \frac{1}{2} \ln(1+t^2)$$

Applying the limits from 0 to 2:

$$I = \left[2 \tan^{-1}(2) - \frac{1}{2} \ln(1+2^2) \right] - \left[0 \cdot \tan^{-1}(0) - \frac{1}{2} \ln(1+0) \right]$$

$$I = 2 \tan^{-1} 2 - \frac{1}{2} \ln 5$$

Question: Let $x_1, x_2, x_3, \dots, x_{10}$ be 10 observations such that $\sum_{i=1}^{10} (x_i - 1)^2 = 90$ and

$\sum_{i=1}^{10} (x_i + 2)^2 = 180$. Then, the standard deviation of $x_1, x_2, x_3, \dots, x_{10}$ is

Options:

- (a) $2\sqrt{2}$
- (b) 3
- (c) 9
- (d) 10

Answer: (b)

Solution:

1. Expand the Given Equations

Let $S_1 = \sum_{i=1}^{10} x_i$ and $S_2 = \sum_{i=1}^{10} x_i^2$. There are $n = 10$ observations.

- From the first summation:

$$\sum_{i=1}^{10} (x_i - 1)^2 = 90$$

$$\sum (x_i^2 - 2x_i + 1) = 90$$

$$S_2 - 2S_1 + 10 = 90 \implies S_2 - 2S_1 = 80 \quad \text{--- (Eq. 1)}$$

- From the second summation:

$$\sum_{i=1}^{10} (x_i + 2)^2 = 180$$

$$\sum (x_i^2 + 4x_i + 4) = 180$$

$$S_2 + 4S_1 + 40 = 180 \implies S_2 + 4S_1 = 140 \quad \text{--- (Eq. 2)}$$

2. Solve for S_1 and S_2

Subtract (Eq. 1) from (Eq. 2) to eliminate S_2 :

$$(S_2 + 4S_1) - (S_2 - 2S_1) = 140 - 80$$

$$6S_1 = 60 \implies S_1 = 10$$

Now substitute $S_1 = 10$ back into (Eq. 1):

$$S_2 - 2(10) = 80$$

$$S_2 - 20 = 80 \implies S_2 = 100$$

3. Calculate Variance and Standard Deviation

The formula for **variance** (σ^2) is:

$$\sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$\sigma^2 = \frac{100}{10} - \left(\frac{10}{10} \right)^2$$

$$\sigma^2 = 10 - (1)^2 = 9$$

The **standard deviation** (σ) is the square root of the variance:

$$\sigma = \sqrt{9} = 3$$

Question: Area bounded by $x = -2y^2$ and $x = 1 - 4y^2$ is
Options:

- (a) $\frac{\sqrt{2}}{3}$ sq.units
- (b) $\frac{2\sqrt{2}}{3}$ sq.units
- (c) $\frac{2}{3}$ sq.units
- (d) $\frac{3\sqrt{3}}{2}$ sq.units

Answer: (b)

Solution:

1. Find the Points of Intersection

We set the two expressions for x equal to each other to find the y -values where the curves meet:

$$-2y^2 = 1 - 4y^2$$

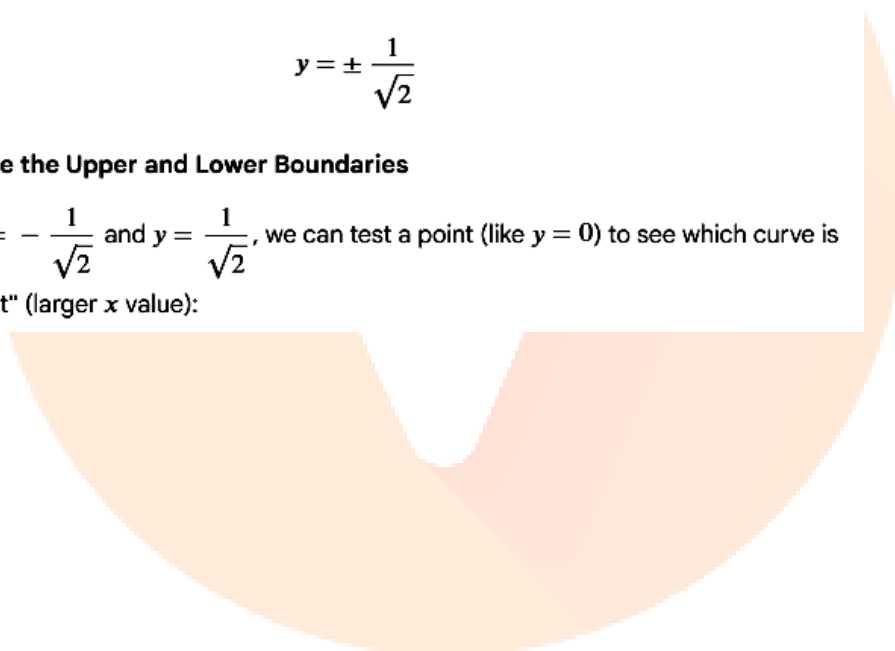
$$4y^2 - 2y^2 = 1$$

$$2y^2 = 1 \implies y^2 = \frac{1}{2}$$

$$y = \pm \frac{1}{\sqrt{2}}$$

2. Determine the Upper and Lower Boundaries

Between $y = -\frac{1}{\sqrt{2}}$ and $y = \frac{1}{\sqrt{2}}$, we can test a point (like $y = 0$) to see which curve is "further right" (larger x value):



- For $x = -2y^2$, if $y = 0$, then $x = 0$.
- For $x = 1 - 4y^2$, if $y = 0$, then $x = 1$.
Since $1 > 0$, the curve $x = 1 - 4y^2$ is the right boundary.

3. Set Up and Evaluate the Integral

The area A is calculated by integrating the difference ($x_{\text{right}} - x_{\text{left}}$) with respect to y :

$$A = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} [(1 - 4y^2) - (-2y^2)] dy$$

$$A = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} (1 - 2y^2) dy$$

Because the integrand is an even function, we can simplify this to:

$$A = 2 \int_0^{1/\sqrt{2}} (1 - 2y^2) dy$$

$$A = 2 \left[y - \frac{2y^3}{3} \right]_0^{1/\sqrt{2}}$$

$$A = 2 \left(\frac{1}{\sqrt{2}} - \frac{2}{3} \cdot \frac{1}{2\sqrt{2}} \right)$$

$$A = 2 \left(\frac{1}{\sqrt{2}} - \frac{1}{3\sqrt{2}} \right) = 2 \left(\frac{3-1}{3\sqrt{2}} \right)$$

$$A = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3} \text{ sq. units}$$

Question: Let $S = \{z \in \mathbb{C} : z^2 + 4z + 16 = 0\}$. Then $\sum_{z \in S} |z + \sqrt{3}i|^2$ is equal to:

Options:

- (a) 23
- (b) 27
- (c) 38
- (d) 42

Answer: (c)

Solution:

1. Solve the Quadratic Equation

The image contains a likely typo: "4x". Given that $z \in \mathbb{C}$ and we are summing over z , the equation is almost certainly meant to be:

$$z^2 + 4z + 16 = 0$$

Using the quadratic formula $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$z = \frac{-4 \pm \sqrt{4^2 - 4(1)(16)}}{2}$$

$$z = \frac{-4 \pm \sqrt{16 - 64}}{2}$$

$$z = \frac{-4 \pm \sqrt{-48}}{2}$$

$$z = \frac{-4 \pm 4\sqrt{3}i}{2}$$

So, the set $\mathcal{S}$ contains two roots:

- $z_1 = -2 + 2\sqrt{3}i$
- $z_2 = -2 - 2\sqrt{3}i$

2. Calculate the Sum of Squared Moduli

We need to find $\sum_{z \in \mathcal{S}} |z + \sqrt{3}i|^2$, which is $|z_1 + \sqrt{3}i|^2 + |z_2 + \sqrt{3}i|^2$.

- For $z_1 = -2 + 2\sqrt{3}i$:

$$z_1 + \sqrt{3}i = (-2 + 2\sqrt{3}i) + \sqrt{3}i = -2 + 3\sqrt{3}i$$

$$|z_1 + \sqrt{3}i|^2 = (-2)^2 + (3\sqrt{3})^2 = 4 + (9 \times 3) = 4 + 27 = 31$$

- For $z_2 = -2 - 2\sqrt{3}i$:

$$z_2 + \sqrt{3}i = (-2 - 2\sqrt{3}i) + \sqrt{3}i = -2 - \sqrt{3}i$$

$$|z_2 + \sqrt{3}i|^2 = (-2)^2 + (-\sqrt{3})^2 = 4 + 3 = 7$$

3. Final Total

Adding the two results together:

$$\text{Sum} = 31 + 7 = 38$$

Question: If the quadratic expression $(\lambda + 2)x^2 - 3\lambda x + 4\lambda = 0$, $\lambda \neq -2$ has two positive roots then the number of possible integral values of λ is

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (b)

Solution:

1. Real Roots (Discriminant $D \geq 0$)

$$D = B^2 - 4AC \geq 0$$

$$(-3\lambda)^2 - 4(\lambda + 2)(4\lambda) \geq 0$$

$$9\lambda^2 - 16\lambda^2 - 32\lambda \geq 0$$

$$-7\lambda^2 - 32\lambda \geq 0$$

Dividing by -1 and reversing the inequality:

$$7\lambda^2 + 32\lambda \leq 0$$

$$\lambda(7\lambda + 32) \leq 0$$

The roots of this expression are 0 and $-\frac{32}{7} \approx -4.57$.

So, $\lambda \in [-\frac{32}{7}, 0]$ (Condition 1).

2. Positive Sum and Product of Roots

For roots α and β to be positive:

- **Sum of roots:** $\alpha + \beta = -\frac{B}{A} = \frac{3\lambda}{\lambda + 2} > 0$
- **Product of roots:** $\alpha\beta = \frac{C}{A} = \frac{4\lambda}{\lambda + 2} > 0$

Both conditions require the rational expression $\frac{\lambda}{\lambda + 2}$ to be positive. Using the number line method (critical points at 0 and -2):

- Positive when $\lambda > 0$ or $\lambda < -2$.
So, $\lambda \in (-\infty, -2) \cup (0, \infty)$ (Condition 2).

3. Finding the Overlap

We now find the intersection of the two regions:

1. $\lambda \in [-4.57, 0]$
2. $\lambda \in (-\infty, -2) \cup (0, \infty)$

The overlapping region is:

$$\lambda \in \left[-\frac{32}{7}, -2\right)$$

4. Identifying Integral Values

The integers within the range $-4.57 \leq \lambda < -2$ are:

- $\lambda = -4$
- $\lambda = -3$

Testing $\lambda = -2$ is excluded by the problem ($\lambda \neq -2$), and $\lambda = 0$ is excluded by the "positive roots" requirement (if $\lambda = 0$, the roots are 0, which is not strictly positive).

There are 2 such integral values.

Question: 3 Numbers are chosen from set of 1,2,3,...,31. Probability that chosen numbers are in AP

Solution:

1. Total Number of Outcomes

The total number of ways to choose 3 numbers from a set of 31 is given by the combination formula $\binom{n}{r}$:

$$\text{Total Outcomes} = \binom{31}{3} = \frac{31 \times 30 \times 29}{3 \times 2 \times 1}$$

$$\text{Total Outcomes} = 31 \times 5 \times 29 = \mathbf{4495}$$

2. Number of Favorable Outcomes (APs)

Let the three chosen numbers be a, b , and c such that $a < b < c$. For them to be in AP, the following condition must be met:

$$a + c = 2b$$

This equation tells us that **$a + c$ must be an even number**. For the sum of two integers to be even, both numbers must have the same parity. This means a and c must be:

- **Both Odd:** Chosen from the set $\{1, 3, 5, \dots, 31\}$. There are **16** odd numbers.

$$\text{Ways} = \binom{16}{2} = \frac{16 \times 15}{2} = 120$$

- **Both Even:** Chosen from the set $\{2, 4, 6, \dots, 30\}$. There are **15** even numbers.

$$\text{Ways} = \binom{15}{2} = \frac{15 \times 14}{2} = 105$$

$$\text{Total Favorable Outcomes} = 120 + 105 = \mathbf{225}$$

3. Final Probability

The probability P is the ratio of favorable outcomes to the total outcomes:

$$P = \frac{225}{4495}$$

Dividing both the numerator and the denominator by 5:

$$P = \frac{45}{899}$$

In decimal form, this is approximately **0.0501** or about a **5%** chance.

Question: Consider a right-angle triangle ABC, inscribed in a parabola $x^2 = 16y$ such that (8, 4) is vertex of the right angle.

The eccentricity of the locus of the centroid of the triangle ABC is

Options:

- (a) 1
- (b) $\frac{1}{\sqrt{2}}$
- (c) $\sqrt{2}$
- (d) $\frac{1}{2}$

Answer: (d)

Solution:

1. Parametric Form of the Parabola

The given parabola is $x^2 = 16y$. This is in the form $x^2 = 4ay$, where $4a = 16$, so $a = 4$.
Any point on this parabola can be represented as:

$$P(t) = (4t, t^2)$$

2. Vertices of the Triangle

Let the vertices of the triangle be A , B , and C .

- The vertex of the right angle is given as $C(8, 4)$.
Checking this point: $8^2 = 16(4)$, so it lies on the parabola. In parametric form $(4t_0, t_0^2)$, we have $4t_0 = 8 \implies t_0 = 2$.
- Let the other two vertices be $A(4t_1, t_1^2)$ and $B(4t_2, t_2^2)$.

3. Right-Angle Condition at C

Since the right angle is at C , the product of the slopes of AC and BC is -1 .

- Slope of $AC = \frac{t_1^2 - 4}{4t_1 - 8} = \frac{(t_1 - 2)(t_1 + 2)}{4(t_1 - 2)} = \frac{t_1 + 2}{4}$
- Slope of $BC = \frac{t_2^2 - 4}{4t_2 - 8} = \frac{t_2 + 2}{4}$

$$\text{Condition: } \frac{t_1 + 2}{4} \cdot \frac{t_2 + 2}{4} = -1$$

$$(t_1 + 2)(t_2 + 2) = -16 \implies t_1 t_2 + 2(t_1 + t_2) + 4 = -16$$

$$t_1 t_2 = -20 - 2(t_1 + t_2) \quad \text{--- (Equation 1)}$$

4. Find the Locus of the Centroid (x, y)

The coordinates of the centroid (x, y) are given by:

$$\bullet \quad x = \frac{4t_1 + 4t_2 + 8}{3} \Rightarrow 3x - 8 = 4(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{3x - 8}{4}$$

$$\bullet \quad y = \frac{t_1^2 + t_2^2 + 4}{3} \Rightarrow 3y - 4 = t_1^2 + t_2^2$$

We know that $t_1^2 + t_2^2 = (t_1 + t_2)^2 - 2t_1t_2$. Substituting Equation 1:

$$3y - 4 = (t_1 + t_2)^2 - 2[-20 - 2(t_1 + t_2)]$$

$$3y - 4 = (t_1 + t_2)^2 + 4(t_1 + t_2) + 40$$

Now, substitute $t_1 + t_2 = \frac{3x - 8}{4}$:

$$3y - 4 = \left(\frac{3x - 8}{4}\right)^2 + 4\left(\frac{3x - 8}{4}\right) + 40$$

$$3y - 4 = \frac{9x^2 - 48x + 64}{16} + 3x - 8 + 40$$

$$3y - 36 - 3x = \frac{9x^2 - 48x + 64}{16}$$

$$16(3y - 3x - 36) = 9x^2 - 48x + 64$$

$$48y - 48x - 576 = 9x^2 - 48x + 64 \Rightarrow 9x^2 = 48y - 640$$

5. Identify the Eccentricity

The final equation $9x^2 = 48y - 640$ is of the form $x^2 = AY + B$, which represents a **parabola**.

By definition, the eccentricity (e) of **any parabola** is always 1.

Question: The maximum value of $E = 16\sin\frac{x}{2}\cos^3\frac{x}{2}$ where $x \in [0, \pi]$, is

Options:

(a) 3

(b) $3\sqrt{3}$

(c) $6\sqrt{3}$

(d) 6

Answer: (b)

Solution:

1. Simplify the Variable

Let $\theta = \frac{x}{2}$.

Given that $x \in [0, \pi]$, the range for θ is $\theta \in [0, \frac{\pi}{2}]$.

The expression now becomes:

$$E = 16 \sin \theta \cos^3 \theta$$

2. Differentiate to Find Critical Points

To find the maximum, we find the derivative $\frac{dE}{d\theta}$ and set it to zero:

$$\frac{dE}{d\theta} = 16 \left[\frac{d}{d\theta} (\sin \theta) \cdot \cos^3 \theta + \sin \theta \cdot \frac{d}{d\theta} (\cos^3 \theta) \right]$$

$$\frac{dE}{d\theta} = 16 [\cos \theta \cdot \cos^3 \theta + \sin \theta \cdot (3 \cos^2 \theta \cdot (-\sin \theta))]$$

$$\frac{dE}{d\theta} = 16 \cos^2 \theta [\cos^2 \theta - 3 \sin^2 \theta]$$

Set $\frac{dE}{d\theta} = 0$:

- $\cos^2 \theta = 0 \implies \theta = \frac{\pi}{2}$ (This gives $E = 0$, which is a minimum).
- $\cos^2 \theta - 3 \sin^2 \theta = 0 \implies \tan^2 \theta = \frac{1}{3}$

Since θ is in the first quadrant, we take the positive root:

$$\tan \theta = \frac{1}{\sqrt{3}} \implies \theta = \frac{\pi}{6} \text{ (or } 30^\circ)$$

3. Calculate the Maximum Value

At $\theta = \frac{\pi}{6}$, we have:

- $\sin \frac{\pi}{6} = \frac{1}{2}$

- $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$

Substitute these into our expression for E :

$$E_{\max} = 16 \left(\frac{1}{2} \right) \left(\frac{\sqrt{3}}{2} \right)^3$$

$$E_{\max} = 16 \cdot \frac{1}{2} \cdot \frac{3\sqrt{3}}{8}$$

$$E_{\max} = \frac{48\sqrt{3}}{16} = 3\sqrt{3}$$

Question: Distance between foci and distance between directrices of hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are 6 and $\frac{8}{3}$ respectively. Find the length of latus rectum.

Solution:

1. Identify the Given Information

For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$:

- **Distance between foci:** $2ae = 6 \implies ae = 3$ --- (Equation 1)
- **Distance between directrices:** $\frac{2a}{e} = \frac{8}{3} \implies \frac{a}{e} = \frac{4}{3}$ --- (Equation 2)

2. Solve for Parameters a and e

Multiply Equation 1 and Equation 2 to eliminate e :

$$(ae) \times \left(\frac{a}{e} \right) = 3 \times \frac{4}{3}$$

$$a^2 = 4 \implies a = 2$$

Now, substitute $a = 2$ into Equation 1 to find the eccentricity (e):

$$2e = 3 \implies e = \frac{3}{2}$$

3. Calculate b^2

For a hyperbola, the relationship between a, b , and e is $b^2 = a^2(e^2 - 1)$.

$$b^2 = 2^2 \left(\left(\frac{3}{2} \right)^2 - 1 \right)$$

$$b^2 = 4 \left(\frac{9}{4} - 1 \right)$$

$$b^2 = 4 \left(\frac{5}{4} \right) = 5$$

4. Find the Length of Latus Rectum

The formula for the length of the latus rectum (LR) is:

$$LR = \frac{2b^2}{a}$$

$$LR = \frac{2(5)}{2}$$

$$LR = 5$$

Question: Let $f(x) = \begin{cases} e^x - 1 & , x < 0 \\ x^2 - 5x + 6 & , x \geq 0 \end{cases}$. If $g(x) = f(|x|) + |f(x)|$. If α and β are the number of points which function $g(x)$ is discontinuous and non-differentiable respectively. Then, the value of $\alpha + \beta$ is

Solution:

1. Analyze the Base Function $f(x)$

The function $f(x)$ is defined piecewise:

- For $x < 0$: $f(x) = e^x - 1$. Note that as $x \rightarrow 0^-$, $f(x) \rightarrow 0$.
- For $x \geq 0$: $f(x) = x^2 - 5x + 6 = (x - 2)(x - 3)$. Note that $f(0) = 6$.

Since the limit from the left (0) and the value at the point (6) are different, **$f(x)$ is discontinuous at $x = 0$.**

2. Define the Function $g(x)$

We evaluate $g(x) = f(|x|) + |f(x)|$ based on different intervals of x :

- **Case 1: $x < 0$**
 - $|x| = -x$ (which is positive), so $f(|x|) = f(-x) = (-x)^2 - 5(-x) + 6 = x^2 + 5x + 6$.
 - For $x < 0$, $e^x < 1$, so $f(x) = e^x - 1$ is always negative. Thus, $|f(x)| = 1 - e^x$.
 - $g(x) = (x^2 + 5x + 6) + (1 - e^x) = x^2 + 5x + 7 - e^x$.
- **Case 2: $x \geq 0$**
 - $|x| = x$, so $f(|x|) = f(x)$. Thus, $g(x) = f(x) + |f(x)|$.
 - If $f(x) \geq 0$ (i.e., $x \in [0, 2] \cup [3, \infty)$), then $g(x) = 2f(x) = 2x^2 - 10x + 12$.
 - If $f(x) < 0$ (i.e., $x \in (2, 3)$), then $g(x) = f(x) - f(x) = 0$.

3. Determine Points of Discontinuity (α)

We check the "junction" points: $x = 0, 2, 3$.

- **At $x = 0$:**
 - LHL = $\lim_{x \rightarrow 0^-} (x^2 + 5x + 7 - e^x) = 0 + 0 + 7 - 1 = 6$.
 - RHL = $\lim_{x \rightarrow 0^+} (2x^2 - 10x + 12) = 12$.

◦ Since $LHL \neq RHL$, $g(x)$ is **discontinuous at $x = 0$** .

- **At $x = 2$ and $x = 3$:** By testing limits, $g(x)$ approaches 0 from both sides at both points. Thus, it is **continuous** at $x = 2$ and $x = 3$.

Therefore, $\alpha = 1$ (point $x = 0$).

4. Determine Points of Non-Differentiability (β)

A point of discontinuity is automatically a point of non-differentiability.

- **At $x = 0$:** Non-differentiable due to discontinuity.

- **At $x = 2$:**

◦ $LHD = \frac{d}{dx}(2x^2 - 10x + 12)$ at $x = 2 \implies 4(2) - 10 = -2$.

◦ $RHD = \frac{d}{dx}(0) = 0$.

◦ Since $LHD \neq RHD$, it is **non-differentiable at $x = 2$** .

- **At $x = 3$:**

◦ $LHD = 0$.

◦ $RHD = \frac{d}{dx}(2x^2 - 10x + 12)$ at $x = 3 \implies 4(3) - 10 = 2$.

◦ Since $LHD \neq RHD$, it is **non-differentiable at $x = 3$** .

Therefore, $\beta = 3$ (points $x = 0, 2, 3$).

5. Final Calculation

The value of $\alpha + \beta$ is:

$$\alpha + \beta = 1 + 3 = 4$$

Question: The shortest distance between the lines

$$\vec{r} = \left(\frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k} \right) + \lambda(2\hat{i} - 5\hat{j} + 6\hat{k})$$

and $\vec{r} = \left(-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k} \right) + \mu(\hat{j} - \hat{k}); \lambda, \mu \in R$ is _____

Solution:

1. Identify the Vectors

From the given equations:

- $\vec{a}_1$ (Position vector of Line 1): $\frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}$
- $\vec{b}_1$ (Direction vector of Line 1): $2\hat{i} - 5\hat{j} + 6\hat{k}$
- $\vec{a}_2$ (Position vector of Line 2): $-\frac{2}{3}\hat{i} + 0\hat{j} - \frac{1}{3}\hat{k}$
- $\vec{b}_2$ (Direction vector of Line 2): $0\hat{i} + \hat{j} - \hat{k}$

2. Perform the Calculations

Step A: Calculate $\vec{a}_2 - \vec{a}_1$

$$\vec{a}_2 - \vec{a}_1 = \left(-\frac{2}{3} - \frac{1}{3}\right)\hat{i} + (0 - 2)\hat{j} + \left(-\frac{1}{3} - \frac{8}{3}\right)\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = -1\hat{i} - 2\hat{j} - 3\hat{k}$$

Step B: Calculate the cross product $\vec{b}_1 \times \vec{b}_2$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -5 & 6 \\ 0 & 1 & -1 \end{vmatrix}$$

$$\vec{b}_1 \times \vec{b}_2 = \hat{i}(5 - 6) - \hat{j}(-2 - 0) + \hat{k}(2 - 0)$$

$$\vec{b}_1 \times \vec{b}_2 = -1\hat{i} + 2\hat{j} + 2\hat{k}$$

Step C: Find the magnitude $|\vec{b}_1 \times \vec{b}_2|$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(-1)^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

Step D: Calculate the dot product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$

$$(-1\hat{i} - 2\hat{j} - 3\hat{k}) \cdot (-1\hat{i} + 2\hat{j} + 2\hat{k}) = (-1)(-1) + (-2)(2) + (-3)(2)$$

$$= 1 - 4 - 6 = -9$$

3. Final Shortest Distance

Substitute the values into the distance formula:

$$d = \frac{|-9|}{3} = \frac{9}{3} = 3$$